

T.F. Notes on Aggregate Transitions using Sequence Space Jacobian (SSJ) Methods

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This document covers the (1) *theory* behind the SSJ solution method (a fast alternative to shooting algorithms or stochastic simulation algorithms) and describes the (2) *numerical* algorithm, pointing to several reference *Jupyter Notebooks*, to implement it for Problem Set 3 in solving aggregate dynamics in Hopenhayn Rogerson.

0. Aiyagari/Krusell Smith Motivation

- Many problems we will look at take this form, $\mathbf{X}_t = (r_t, w_t)$

$$\begin{aligned} V_t(e, k_- | \mathbf{X}_t) &= \max_{c, k} u(c) + \beta \sum_{e'} V_{t+1}(e', k, \mathbf{X}_{t+1}) P(e, e') \\ \text{s.t. } c + k &= (1 + r_t) k_- + w_t e l, \quad k \geq 0 \end{aligned}$$

- Where the law of motion of the distribution of people waking up at t with $D_t(e, K_-) \equiv \Pr(e_t = e, k_{t-1} \in K_-)$

$$D_{t+1}(e', K) = \sum_e D_t(e, k_-) P(e, e') \approx \sum_e \int_{k_-} P(e, e') \omega(k_t^*(e, k_-) \approx K) D_t(e, k_-)$$

- At the steady state if you start there, you stay there: $D_0 = D_{ss}$.
- Get policies $c_t^*(e, k_-)$ and $k_t^*(e, k_-)$ as a function of the future path $\{r_s, w_s\}_{s \geq t}$. Thus, $D_t(e', K)$ depends also on the path of these aggregates.
- Aggregate capital holdings supplied by households

$$\mathcal{K}_t(\{r_s, w_s\}) = \sum_e \int_{k_-} k_t^*(e, k_-) D_t(e, dk_-)$$

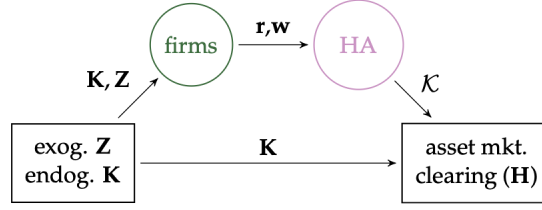
- Producers demand capital and labor

$$Y_t = Z_t K_{t-1}^\alpha L_t^{1-\alpha}, \quad r_t = \alpha Z_t \left(\frac{K_{t-1}}{L_t} \right)^{\alpha-1} - \delta, \quad w_t = (1 - \alpha) Z_t \left(\frac{K_{t-1}}{L_t} \right)^\alpha$$

- We have exogenous $\{Z_t, L_t = \sum \pi(e) e l\}$ and endogenous $\{r_t, w_t\}$ and $\{K_t\}$. For asset markets to clear, the amount of production capital (and demanded by firms) the economy wakes up with tomorrow needs to be the same that households want to save today given current and future prices. Goods market clearing follows from Walras Law after asset market clearing

$$H_t(\mathbf{K}, \mathbf{Z}) \equiv \mathcal{K}_t(\{r_s, w_s\}) - K_t = \mathcal{K}_t \left(\left\{ \alpha Z_s \left(\frac{K_{s-1}}{\sum \pi(e) e l} \right)^{\alpha-1} - \delta, (1 - \alpha) Z_s \left(\frac{K_{s-1}}{\sum \pi(e) e l} \right)^\alpha \right\} \right) - K_t = 0$$

Figure 3: DAG representation of Krusell-Smith economy

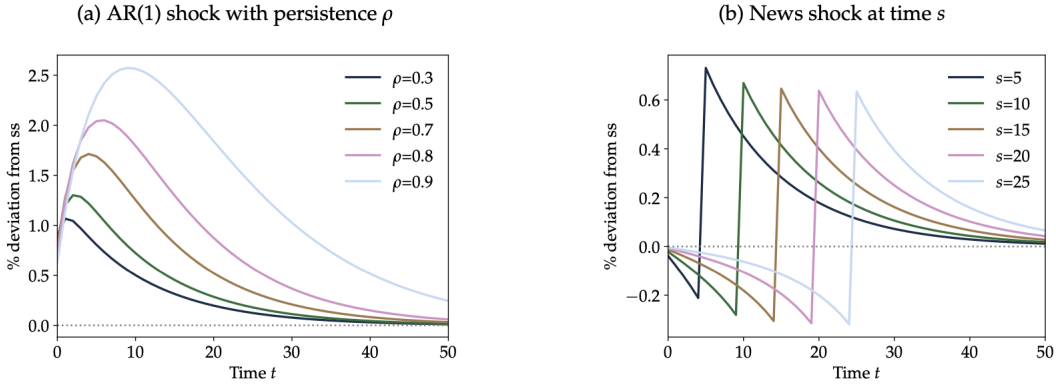


- Two key objects we care about: $\partial \mathcal{K} / \partial r = \mathcal{J}^{\mathcal{K}, r}$ and $\partial \mathcal{K} / \partial w = \mathcal{J}^{\mathcal{K}, w}$ that give rise to

$$[\mathbf{H}_{\mathbf{K}}]_{t,s} = \frac{\partial \mathcal{K}_t}{\partial r_{s+1}} \frac{\partial r_{s+1}}{\partial K_s} + \frac{\partial \mathcal{K}_t}{\partial w_{s+1}} \frac{\partial w_{s+1}}{\partial K_s} - 1_{\{s=t\}}$$

- When the shock takes place at date $s = 0$, households are surprised by a higher return on existing assets r_0 . They save some of this additional return (**a standard wealth effect**), accumulating assets that they later progressively de-cumulate.
- When the shock takes place at later dates $s > 0$, households also have an **inter-temporal substitution response**, which leads them to save in anticipation of the increase in r . This generates a “tent” pattern in the Jacobian

Figure 1: Impulse responses of capital to 1% TFP shocks in the Krusell-Smith model



I. Heterogeneous Agent Model: Motivation and General Formulations

Endogenous v.s. Exogenous Formulation.

- How do we compute IRF of a model which is reduced to a system of non-linear equations?

$$\mathbf{H}(\mathbf{U}, \mathbf{Z}) = 0$$

- $\mathbf{Z}$ is $\mathbf{Z}_t$ $n_z \times 1$ vector of *shocks* and $\mathbf{U}$ is $\mathbf{U}_t$ $n_u \times 1$ vector of *unknowns* for every t stacked.
- $\mathbf{H}(\mathbf{U}, \mathbf{Z}) = 0$ is a system of $n_u \times T$ equations in $n_u \times T$ unknowns.
- Applying the implicit function theorem we solve for path of $\mathbf{U}$
- Apply chain rule to obtain IRFs for any endogenous aggregate

$$\mathbf{H}_U d\mathbf{U} + \mathbf{H}_Z d\mathbf{Z} = 0 \implies d\mathbf{U} = -\mathbf{H}_U^{-1} \mathbf{H}_Z d\mathbf{Z}$$

$$\mathbf{Y} = \mathbf{Y}(\mathbf{U}, \mathbf{Z}) \implies d\mathbf{Y} = \mathbf{Y}_U d\mathbf{U} + \mathbf{Y}_Z d\mathbf{Z} = \left[\mathbf{Y}_Z - \mathbf{Y}_U \mathbf{H}_U^{-1} \mathbf{H}_Z \right] d\mathbf{Z}$$

- Two problems: (1) how to compute Jacobians inside of $\mathbf{H}_x$ and (2) how to reduce dimensionality, since $\mathbf{H}_U^{-1}$ $n_u T \times n_u T$ with a lot of n_u difficult to invert.
- We tackle (1) first and (2) via examples.

Inputs v.s. Outputs Formulation.

- A het. agent model maps n_x inputs to n_y outputs, $\mathbf{X}_t$ (e.g. r and w) $\rightarrow \mathbf{Y}_t$ (e.g. K, L)
- The dynamic model and its steady state fixed point $(\mathbf{Y}_{ss}, \mathbf{v}_{ss}, \mathbf{D}_{ss}, \mathbf{X}_{ss})$ is defined by

$$\begin{aligned} \mathbf{v}_t &= v(\mathbf{v}_{t+1}, \mathbf{X}_t) & \mathbf{v}_{ss} &= v(\mathbf{v}_{ss}, \mathbf{X}_{ss}) \\ \mathbf{D}_{t+1} &= \Lambda(\mathbf{v}_{t+1}, \mathbf{X}_t)' \mathbf{D}_t = \Lambda_t' \mathbf{D}_t & \rightarrow & \mathbf{D}_{ss} = \Lambda(\mathbf{v}_{ss}, \mathbf{X}_{ss})' \mathbf{D}_{ss} = \Lambda_{ss}' \mathbf{D}_{ss} \\ \mathbf{Y}_t &= y(\mathbf{v}_{t+1}, \mathbf{X}_t)' \mathbf{D}_t = \mathbf{y}_t' \mathbf{D}_t & \mathbf{Y}_{ss} &= y(\mathbf{v}_{ss}, \mathbf{X}_{ss})' \mathbf{D}_{ss} = \mathbf{y}_{ss}' \mathbf{D}_{ss} \end{aligned}$$

- Distributions $\mathbf{D}_t$ and values $\mathbf{v}_t$ $n_g \times 1$ are discretized over n_g grid points.
- Importantly $\mathbf{v}$ does not directly depend on $\mathbf{D}$ only on aggregate general equilibrium inputs $\mathbf{X}_t$ that intermediate the distribution of agents and outputs. To see an extension where it does, see Masao Fukui's (MIT JMC) job market paper.
- Key problem: $\mathbf{v}$ and $\mathbf{D}$ are defined *recursively*.
- Λ_t is an $n_g \times n_g$ is a transition matrix that depends on agent's policies, and thus, on $y(\mathbf{v}_{t+1}, \mathbf{X}_t)$, or $\mathbf{y}_t$, an $n_g \times n_y$ matrix for all outcomes we care about (e.g. consumption, savings, employment).
- Given $\mathbf{D}_0 = \mathbf{D}_{ss}$, and $\mathbf{X}_{T-1} = \mathbf{X}_{ss}$, and $\mathbf{v}_T = \mathbf{v}_{ss}$, we can stack the model over time as $T \times n_x$ inputs $\mathbf{X}$ and $T \times n_y$ outputs $\mathbf{Y}$.

$$\mathbf{Y} = h(\mathbf{X})$$

- The Jacobians $\mathcal{J}$ of all outputs in $\mathbf{Y}$ and for all inputs in $\mathbf{X}$ are all we need (i.e. are sufficient statistics) to locally (i.e. to a first-order) describe this mapping.
- Assume only one input and one output $n_x = n_y = 1$ for clarity (we extend later) so that $\mathbf{X}$ and $\mathbf{Y}$ are T -dimensional vectors.
- Our goal is to perturb this system around the steady state by dx . Then, one approach is to compute¹

$$\mathcal{J}_{t,s} = \frac{dY_t^s}{dx} = \left(\frac{Y_t^s - Y_{ss}}{dx} \right)$$

- This is the sth column of the Jacobian, a numerical version of $\lim_{dx \rightarrow 0} \frac{dY^s}{dx}$, or the IRF of the economy to current/news shock at s for all $t = 0, \dots, T$.
- Let $\mathbf{e}^s$ $T \times 1$ vector 0's everywhere except at the s 'th entry, where it has a 1.
- A "shock at time s " is when the $T \times 1$ input vector $\mathbf{X}$ is given by the steady state vector except for a change in the s 'th entry.

$$\mathbf{X}^s \equiv \mathbf{X}_{ss} + \mathbf{e}^s dx$$

- Importantly: this is a one-time completely transitory and unexpected "MIT shock"
- If $s > 0$ then this is a *news* shock, if $s = 0$ then this is a one-time *current* shock.
- $\mathbf{v}_t^s, \mathbf{D}_t^s$, and Y_t^s solve the dynamic equilibrium for such shocked $\mathbf{X}^s$ input vector
- This notation reflects functions at time t in response to the shock at time s
- Backward iteration will give you the transition matrix between states at time t (given shock in s) implied by the optimal policy given such perturbed $\mathbf{X}^s$ at the solution $\mathbf{v}_{t+1}^s$

$$\mathbf{y}_t^s \equiv y(\mathbf{v}_{t+1}^s, X_t^s) \rightarrow \Lambda_t^s \equiv \Lambda(\mathbf{v}_{t+1}^s, X_t^s)$$

- We look at changes relative to the steady state

$$dY_t^s \equiv Y_t^s - Y_{ss}, \quad d\mathbf{y}_t^s \equiv \mathbf{y}_t^s - \mathbf{y}_{ss}, \quad d\Lambda_t^s \equiv \Lambda_t^s - \Lambda_{ss}, \quad d\mathbf{D}_t^s \equiv \mathbf{D}_t^s - \mathbf{D}_{ss}$$

II. Direct Method.

¹In practice two-sided differentiation works better letting $\mathcal{J}_{\cdot,s} = \frac{Y^{s+} - Y^{s-}}{2dx}$ where the output responses are

$$\mathbf{Y}^{s+} \text{ to } \mathbf{X}^{s+} \equiv \mathbf{X}_{ss} + \mathbf{e}^s dx, \quad \mathbf{Y}^{s-} \text{ to } \mathbf{X}^{s-} \equiv \mathbf{X}_{ss} - \mathbf{e}^s dx$$

Also, Python's automatic differentiation *jax* package has the benefit of computing the exact derivatives without approximation error, but is hard to work with.

- Take $\mathbf{X}^s$ path as given.
- Iterate backwards $\mathbf{v}_T = \mathbf{v}_{ss}$ and compute the path of $\mathbf{v}_t^s$ for that particular shock and the associated transition matrix Λ_t^s and individual policies $\mathbf{y}_t^s$ for $t = T - 1, \dots, 0$.
- Iterate $\mathbf{D}_0 = \mathbf{D}_{ss}$ forward applying the transition matrices to solve the distributions $\mathbf{D}_t^s$ for $t = 1, \dots, T - 1$
- At each t take $Y_t^s = (\mathbf{y}_t^s)' \mathbf{D}_t^s$, and compute numerical derivatives $\mathcal{J}_{t,s} = \frac{dY_t^s}{dx}$ as above
- Repeat for each of the columns $s = 0, T - 1$, calculating the IRF for the input shock in the model

III. Fake News Algorithm.

- Start from two key objects of interest describing (1) aggregation and (2) distributional transition

$$Y_t^s = (\mathbf{y}_t^s)' \mathbf{D}_t^s, \quad \mathbf{D}_{t+1}^s = (\Lambda_t^s)' \mathbf{D}_t^s$$

Lemma 1. For any $s \geq 1, t \geq 1$, given a shock s , **all that matters is the distance** between s and t .

$$\mathbf{y}_t^s = \begin{cases} \mathbf{y}_{ss} & s < t \\ \mathbf{y}_{T-1-(s-t)}^{T-1} & s \geq t \end{cases} \quad \text{and} \quad \Lambda_t^s = \begin{cases} \Lambda_{ss} & s < t \\ \Lambda_{T-1-(s-t)}^{T-1} & s \geq t \end{cases}$$

- Let that sink in. A shock at s ahead of t is equivalent to a shock at $T - 1$ ahead of $T - 1 - (s - t)$. Agents only care about the distance to the shock $s - t$, rather than calendar time t and s separately, when deciding on their own behavior. The response of their consumption policy at time t to any shock at time $t + 1$ is the same as the response of their consumption policy at time 0 to the same shock at time 1
- But then, we can compute *all* policies we need $\mathbf{y}_t^s$ (and all transition matrices Λ_t^s) for any shock dates s and any t from a single perturbation of the input at date $s = T - 1$.
- Simply truncate where you want the distance to be: letting the perturbation happen at $s = T - 1$ going backwards *once* gives us *all* possible distances $s - t$ we will need.
- This replaces the T backward iterations (one for each s) of the direct algorithm by a single backward iteration starting from a shock at date $T - 1$.
- Another crucial issue is that the transition begins and ends in the same steady state (no permanent shock) so that $\mathbf{D}_0 = \mathbf{D}_{ss}$ and $\mathbf{D}_{T-1} = \mathbf{D}_{ss}$ ever after.
- For any $s \geq 1, t \geq 1$, let the **(recursive) fake news matrix**

$$\mathcal{F}_{t,s} \cdot dx \equiv dY_t^s - dY_{t-1}^{s-1} = \left(\mathcal{J}_t^s - \mathcal{J}_{t-1}^{s-1} \right) dx$$

- Captures the difference between the aggregate response of the output at t to a shock at date s , and its response at $t - 1$ to a shock at date $s - 1$.
- The fake news matrix is essentially subtracting two cols of the Jacobian dY_t^s (think about how one solves finite sequences subtracting two shifted infinite sequences)
- This will be useful later, since we will build Jacobians recursively $\mathcal{J}_t^s = \mathcal{F}_{t,s} + \mathcal{J}_{t-1}^{s-1}$
- Since policies (and the transition matrix) only depends on the distance between t and s , by **Lemma 1**, $\mathbf{y}_t^s = \mathbf{y}_{T-1-(s-t)}^{T-1} = \mathbf{y}_{t-1}^{s-1}$ since they have the **same distance**

$$Y_t^s = (\mathbf{y}_t^s)' \mathbf{D}_t^s, \quad Y_t^s - Y_{t-1}^{s-1} = (\mathbf{y}_t^s)' [\mathbf{D}_t^s - \mathbf{D}_{t-1}^{s-1}]$$

Lemma 2. Assume $\mathbf{D}_0 = \mathbf{D}_{ss}$. For infinitesimal dx we have that for $s \geq 1, t \geq 1$

$$\mathcal{F}_{t,s} \cdot dx = dY_t^s - dY_{t-1}^{s-1} = \mathbf{y}_{ss}' (\Lambda_{ss}')^{t-1} d\mathbf{D}_1^s$$

- **Proof.** Linearize around the steady state without shocks $Y_t = Y_{ss}, \mathbf{y}_t^s = \mathbf{y}_{ss}, \mathbf{D}_t^s = \mathbf{D}_{ss}$ as $dx \rightarrow 0$

1. In the limit (around the steady state) we have that differentiating via the chain rule we get that in het agent models to a first order, changes in aggregates can reflect both changes in policies and in the distribution.

$$Y_t^s = (\mathbf{y}_t^s)' \mathbf{D}_t^s \implies dY_t^s = \mathbf{y}_{ss}' d\mathbf{D}_t^s + (d\mathbf{y}_t^s)' \mathbf{D}_{ss}$$

2. Around the steady state, we can approximate aggregate changes for any t, s

$$dY_t^s = \mathbf{y}_{ss}' d\mathbf{D}_t^s + (d\mathbf{y}_t^s)' \mathbf{D}_{ss}, \quad dY_{t-1}^{s-1} = \mathbf{y}_{ss}' d\mathbf{D}_{t-1}^{s-1} + (d\mathbf{y}_{t-1}^{s-1})' \mathbf{D}_{ss}$$

3. Subtracting the two we get that

$$dY_t^s - dY_{t-1}^{s-1} = \mathbf{y}_{ss}' (d\mathbf{D}_t^s - \mathbf{D}_{t-1}^{s-1}) + (d\mathbf{y}_t^s - d\mathbf{y}_{t-1}^{s-1})' \mathbf{D}_{ss}$$

4. (**Lemma 1**) Since only the relative distance matters for DP policies, when comparing two aggregate changes shifted in time but separated by the same distance, only distributional dynamics show up (not policies).

$$\mathbf{y}_t^s = \mathbf{y}_{T-1-(s-t)}^{T-1} = \mathbf{y}_{t-1}^{s-1} \implies dY_t^s - dY_{t-1}^{s-1} = \mathbf{y}_{ss}' (d\mathbf{D}_t^s - \mathbf{D}_{t-1}^{s-1}) + 0$$

5. Defining the fake news matrix with entries

$$\mathcal{F}_{t,s} \cdot dx = dY_t^s - dY_{t-1}^{s-1} = \mathbf{y}_{ss}' (d\mathbf{D}_t^s - d\mathbf{D}_{t-1}^{s-1})$$

6. Re-express $d\mathbf{D}_t^s$ and $d\mathbf{D}_{t-1}^{s-1}$ by taking derivatives of the law of motion around the steady state

$$\mathbf{D}_{t+1}^s = (\Lambda_t^s)' \mathbf{D}_t^s \implies d\mathbf{D}_{t+1}^s = (d\Lambda_t^s)' \mathbf{D}_{ss} + (\Lambda_{ss})' d\mathbf{D}_t^s, \quad d\mathbf{D}_{t-1}^{s-1} = (d\Lambda_{t-2}^{s-1})' \mathbf{D}_{ss} + (\Lambda_{ss})' d\mathbf{D}_{t-2}^{s-1}$$

7. Changes in the distribution can come from changes in the policy-induced weights over time as given by policies and shocks, and the recursive component of the distribution we carry over.

$$d\mathbf{D}_t^s - d\mathbf{D}_{t-1}^{s-1} = (d\Lambda_{t-1}^s - d\Lambda_{t-2}^{s-1})' \mathbf{D}_{ss} + (\Lambda_{ss})' (d\mathbf{D}_{t-1}^s - d\mathbf{D}_{t-2}^{s-1})$$

8. (**Lemma 1**) Again, since the policies are the same given the same distance

$$d\Lambda_{t-1}^s = d\Lambda_{t-2}^{s-1} \implies d\mathbf{D}_t^s - d\mathbf{D}_{t-1}^{s-1} = (\Lambda_{ss})' (d\mathbf{D}_{t-1}^s - d\mathbf{D}_{t-2}^{s-1})$$

9. We then get that, to a first order around the steady state, all we need to do is apply Λ_{ss} from the steady state policy functions

$$d\mathbf{D}_t^s - d\mathbf{D}_{t-1}^{s-1} = (\Lambda_{ss})' (\Lambda_{ss})' (d\mathbf{D}_{t-2}^s - d\mathbf{D}_{t-3}^{s-1}) = (\Lambda_{ss}')^2 (d\mathbf{D}_{t-2}^s - d\mathbf{D}_{t-3}^{s-1}) = \dots = (\Lambda_{ss}')^{t-1} (d\mathbf{D}_1^s - d\mathbf{D}_0^{s-1})$$

10. Since $\mathbf{D}_0 = \mathbf{D}_{ss}$ as a starting point, we have that

$$d\mathbf{D}_0^{s-1} = 0 \quad \forall s \geq 1 \implies d\mathbf{D}_t^s - d\mathbf{D}_{t-1}^{s-1} = (\Lambda_{ss}')^{t-1} d\mathbf{D}_1^s$$

Concluding our proof that this

$$\mathcal{F}_{t,s} \cdot dx = dY_t^s - dY_{t-1}^{s-1} = \mathbf{y}_{ss}' (d\mathbf{D}_t^s - d\mathbf{D}_{t-1}^{s-1}) = \mathbf{y}_{ss}' (\Lambda_{ss}')^{t-1} d\mathbf{D}_1^s$$

- Suppose we know the path of the aggregate output $Y_t \forall t = 0, \dots$ in response to a shock at date $s - 1$. How does this compare to the path of Y_t to a shock on date s from $t = 1, \dots$ onwards where the distance is the same, but the only difference is that the path is shifted by one?
- By Lemma 1 the behavior of agents at all dates is identical in both cases. The only difference is that the initial distribution in the second case is starting from $\mathbf{D}_1^s$ rather than $\mathbf{D}_{ss}$ which already encapsulates the fact that agents learned about the shock and re-optimized on impact.
- This difference in initial distribution affects aggregates at all dates as if agents followed their steady state behavior
- For a given s $\mathcal{F}_{t,s}$ can be interpreted as the impulse response to a “date- s fake news shock”. This is a shock to date s announced at date 0 and retracted at date 1. At date 0, agents react to the announcement, which leads to the distribution $d\mathbf{D}_1^s$. After the announcement

is retracted, they *revert back to steady state policies*, so the effect on output at all dates $t \geq 1$ is just $\mathbf{y}'_{ss} \cdot (\Lambda'_{ss})^{t-1} d\mathbf{D}_1^s$.

Lemma 3. Let the *expectations vector* $\mathcal{E}_t$ $n_g \times 1$ (this is known as the observability matrix) for outcome Y at time t , which can be solved recursively and is a function of *steady-state* objects only

$$\mathcal{E}_t \equiv (\Lambda_{ss})^t \mathbf{y}_{ss} \implies \mathcal{E}_t = \Lambda_{ss} \mathcal{E}_{t-1}, \quad \mathcal{E}_0 = \mathbf{y}_{ss}$$

Then we can express entries of the date- s fake news shock matrix as

$$\mathcal{F}_{t,s} \cdot dx = dY_t^s - dY_{t-1}^{s-1} = \left((\Lambda_{ss})^{t-1} \mathbf{y}_{ss} \right)' d\mathbf{D}_1^s = \mathcal{E}'_{t-1} d\mathbf{D}_1^s$$

- Recall that transitions v.s. expectations v.s. aggregation always take the following form:

$$\mathbf{d}_1 = \mathbf{T}' \mathbf{d}_0, \quad Y_0 = \mathbf{y}'_0 \mathbf{d}_0, \quad \mathbf{y}_1 = \mathbf{T} \mathbf{y}_0, \quad \mathbf{y}_t = \mathbf{T}^t \mathbf{y}_0$$

- For each grid point in $\mathcal{E}_t$, the time path of $\mathcal{E}_t$ represents the expected time path of outcome Y , in the steady state, for an agent starting on that grid-point.

Proposition 1. Define the **fake-news matrix** such that on impact it's just the aggregate response (since at $t = 0$ $dY_{-1}^{s-1} = Y_{ss} - Y_{ss}$), while for news is just fake news expression we derived above

$$\mathcal{F}_{t,s} \cdot dx = dY_t^s - dY_{t-1}^{s-1} \equiv \begin{cases} dY_0^s & t = 0 \\ \mathcal{E}'_{t-1} d\mathbf{D}_1^s & t \geq 1 \end{cases} \iff \mathcal{F}_{t,s} = \mathcal{J}_t^s - \mathcal{J}_{t-1}^{s-1}$$

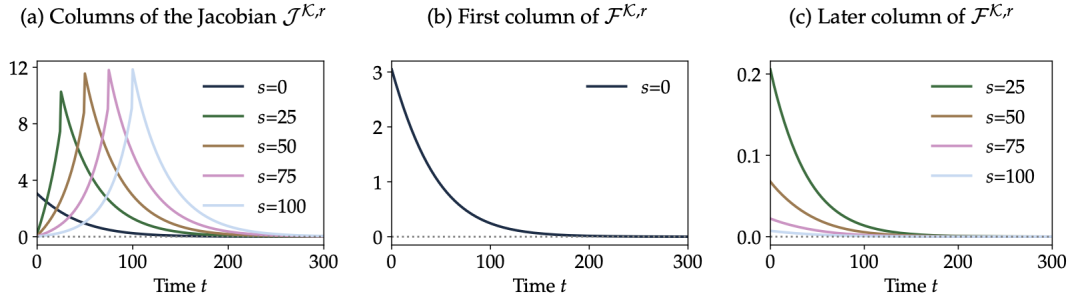
Instead, we are after the Jacobian, which can be expressed recursively

$$\mathcal{J}_{t,s} = \frac{dY_t^s}{dx} = \frac{(Y_t^s - Y_{ss})}{dx} \iff \mathcal{J}_{t,s} dx = dY_t^s$$

Either compactly as $\mathcal{J}_{t,s} = \sum_{k=0}^{\min\{s,t\}} \mathcal{F}_{t-k,s-k}$ or explicitly as

$$\begin{aligned} \mathcal{J}_{t,s} &= \mathcal{F}_{t,s} = \frac{dY_t^s}{dx} - 0 \text{ for } t = 0 \text{ or } s = 0 \\ \mathcal{J}_{t,s} &= \mathcal{J}_{t-1,s-1} + \mathcal{F}_{t,s} \text{ for } t, s \geq 1 \end{aligned}$$

Figure 2: Jacobian $\mathcal{J}^{\mathcal{K},r}$ and fake news matrix $\mathcal{F}^{\mathcal{K},r}$ in the Krusell-Smith model.



- Cool fact: first column (row) of $\mathcal{J}$ is the same as the first column (row) of $\mathcal{F}$.
- Later columns of the fake news matrix: By the “fake news” interpretation, these columns represent the behavior of aggregate assets when households first save at date 0 in anticipation of an increase in r at date s , and then dis-save after the announcement is retracted at date 1.
- Note that the s -th column is the IRF to a news shock about an impulse in x_s starting from the POV of t and plotting the IRF forward.
- Cool intuition: The impulse responses to AR(1) TFP paths of persistence ρ can be reinterpreted as the effect of the simultaneous news, at date 0, of an increase of ρ^s in TFP at times $s = 0, 1, \dots$

Proof.

1. When $s, t \geq 1$ the recursion follows from **Lemma 1**.
2. When $t = 0, s \geq 1$, since there's no change in the ss, **the Jacobian and the fake news matrix have the same first row**

$$\mathcal{J}_{t,s} \cdot dx = \mathcal{F}_{t,s} \cdot dx = dY_0^s$$

3. When $s = 0, t \geq 1$, the economy is back in the steady state at t , and thus, no effects of the policies in the future $dy_t^0 = 0$, by **Lemma 1**.

$$Y_t^0 = (y_t^0)' D_t^0 \implies dY_t^0 = y_{ss}' dD_t^0 + (dy_t^0)' D_s$$

4. All the effects happen via the distribution

$$\mathcal{J}_{t,0} \cdot dx = dY_t^0 = y_{ss}' dD_t^0$$

5. Since $d\Lambda_t^0 = 0$ for all $t \geq 1$ (as the policies do not change when we get back to the steady state), we can write applying our **lemma** that

$$\mathcal{J}_{t,0} \cdot dx = y_{ss}' dD_t^0 = y_{ss}' \Lambda_{ss}' dD_{t-1}^0 = \dots = y_{ss}' (\Lambda_{ss}')^{t-1} dD_1^0 = \mathcal{E}_{t-1}' dD_1^0$$

6. So the IRF for a current shock $s = 0$ in the Jacobian matrix can be traced out simply by

$$\mathcal{J}_{t,0} \cdot dx = \mathcal{E}_{t-1}' dD_1^0 = \mathcal{F}_{t,0} \cdot dx$$

7. That is, the first column of the Jacobian and the fake-news matrix are the same: both are responses to one time shocks at $s = 0$.

- **Proposition 1** implies that to solve for the response of heterogeneous agents to changes in their environment at any date s we only need to know two *on impact* changes and one SS object

- $\mathcal{E}_t$ expectations vector that can be obtained iterating on SS objects
- dY_0^s : the aggregate *initial* response $t = 0$ to shocks at date s
- $d\mathbf{D}_1^s$ (the response of the distribution at date 1 to shocks at date s , since at 0 the distribution is stuck and thus does not change

$$dY_0^s = (d\mathbf{y}_0^s)' \mathbf{D}_{ss}, \quad d\mathbf{D}_1^s = (d\Lambda_0^s)' \mathbf{D}_{ss}$$

- Then, it is possible to project the effect at any date t from knowledge of the effects of future shocks on aggregates dY_0^s and distributions $d\mathbf{D}_1^s$ at date 0.

Algorithm for a single input and output.

0. Compute the steady state of the model.

1. Calculate the $(T - 1) n_g \times 1$ expectation vectors $\mathcal{E}_t \equiv (\Lambda_{ss})^t \mathbf{y}_{ss}$ from the recursion

$$\mathcal{E}_t = \Lambda_{ss} \mathcal{E}_{t-1}, \quad \mathcal{E}_0 = \mathbf{y}_{ss}$$

2. Get $\mathbf{y}_0^s$ and Λ_0^s as well as dY_0^s and $d\mathbf{D}_1^s$. All we need is to solve one backward iteration for time $t = 0$ response to shocks at $s \leq T - 1$, and get how policies and transition matrices change relative to the steady state ones.

2.1 For each $s = 0, \dots, T - 1$ use a **single backward iteration from time $T - 1$** to get $\mathbf{y}_0^s$ and Λ_0^s

2.2 Combine them with the initial steady state distribution $\mathbf{D}_{ss}$ to form T scalars (the effect on the output at date 0 from the shock to the input at date s) using $d\mathbf{y}_0^s = \mathbf{y}_0^s - \mathbf{y}_{ss}$ from the backward iteration

$$\mathcal{Y}_s dx \equiv dY_0^s = (d\mathbf{y}_0^s)' \mathbf{D}_{ss}$$

2.3 Get also the $T n_g \times 1$ vectors with the change in the distribution at date 1 from the shock at date s using $d\Lambda_0^s = \Lambda_0^s - \Lambda_{ss}$ from the backward iteration²

$$\mathcal{D}_s dx \equiv d\mathbf{D}_1^s = (d\Lambda_0^s)' \mathbf{D}_{ss}$$

3. Form the fake-news matrix $\mathcal{F}$. Here we simply apply the **Proposition** used to trace out the fake news IRF (impulse starting at 0 of a shock at some s such that agents are told in the second period that it was fake, and thus, revert back to SS policies, but the distribution keeps moving because of the initial change).

$$\mathcal{F}_{t,s} \cdot dx = dY_t^s - dY_{t-1}^{s-1} \equiv \begin{cases} dY_0^s & t = 0 \\ \mathcal{E}_{t-1}' d\mathbf{D}_1^s & t \geq 1 \end{cases}$$

²In practice, it is usually more accurate to compute the differences $d\mathbf{y}_0^s$ and $d\Lambda_0^s$ by subtracting “ghost runs” rather than steady states. That is, compute $\mathbf{y}_0^s$ for some small dx . Repeat with $dx = 0$ to get $\tilde{\mathbf{y}}_0^s$. Set $d\mathbf{y}_0^s = \mathbf{y}_0^s - \tilde{\mathbf{y}}_0^s$. This procedure is more accurate than subtracting steady state values whenever those have not fully converged $\tilde{\mathbf{y}}_0^s \neq \mathbf{y}_{ss}$. Do the same for $d\Lambda_0^s$.

3.1 The first row $t = 0$ simply contains $\mathcal{Y}_s \equiv \frac{dY_0^s}{dx} = \left(\frac{dy_0^s}{dx}\right)' \mathbf{D}_{ss}, \quad \forall s$

3.2 Other rows $t \geq 1$ contain $\mathcal{E}'_{t-1} \mathcal{D}_s, \quad \mathcal{D}_s \equiv \frac{d\mathbf{D}_1^s}{dx} = \left(\frac{d\Lambda_0^s}{dx}\right)' \mathbf{D}_{ss}$

$$\mathcal{F} = \begin{bmatrix} \mathcal{Y}_0 & \mathcal{Y}_1 & \mathcal{Y}_2 & \dots & \mathcal{Y}_{T-1} \\ \mathcal{E}'_0 \mathcal{D}_0 & \mathcal{E}'_0 \mathcal{D}_1 & \mathcal{E}'_0 \mathcal{D}_2 & & \mathcal{E}'_0 \mathcal{D}_{T-1} \\ \mathcal{E}'_1 \mathcal{D}_0 & \mathcal{E}'_1 \mathcal{D}_1 & \mathcal{E}'_1 \mathcal{D}_2 & & \mathcal{E}'_1 \mathcal{D}_{T-1} \\ \vdots & \vdots & \vdots & & \vdots \\ \mathcal{E}'_{T-2} \mathcal{D}_0 & \mathcal{E}'_{T-2} \mathcal{D}_1 & \mathcal{E}'_{T-2} \mathcal{D}_2 & \dots & \mathcal{E}'_{T-2} \mathcal{D}_{T-1} \end{bmatrix}$$

4. Apply the proposition again to build the Jacobian recursively

$$\mathcal{J}_{t,s} = \mathcal{J}_{t-1,s-1} + \mathcal{F}_{t,s}, \quad \mathcal{J}_{t,s} = \sum_{k=0}^{\min\{s,t\}} \mathcal{F}_{t-k,s-k}$$

- This implies that the element (t, s) of the Jacobian $\mathcal{J}$ is the sum of the (t, s) element of the $\mathcal{F}$ matrix and of all the elements on the diagonal to its immediate upper left.
- $\mathcal{J}_{1,1}$ example and $\mathcal{J}_{x,0}$ example (IRF decay for current shocks)

$$\mathcal{J} = \begin{bmatrix} \mathcal{Y}_0 & \mathcal{Y}_1 & \mathcal{Y}_2 & \dots & \mathcal{Y}_{T-1} \\ \vdots & \mathcal{E}'_0 \mathcal{D}_1 + \mathcal{Y}_0 & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}, \quad \mathcal{J} = \begin{bmatrix} \mathcal{Y}_0 & \mathcal{Y}_1 & \mathcal{Y}_2 & \dots & \mathcal{Y}_{T-1} \\ \mathcal{E}'_0 \mathcal{D}_0 & \mathcal{E}'_0 \mathcal{D}_1 + \mathcal{Y}_0 & & & \\ \mathcal{E}'_1 \mathcal{D}_0 & & & & \\ \vdots & \vdots & \vdots & & \vdots \end{bmatrix}$$

- $\mathcal{J}_{2,2}$ example, $\mathcal{J}_{2,1}$ and $\mathcal{J}_{1,2}$ example (tent shape for news shocks)

$$\mathcal{J} = \begin{bmatrix} \mathcal{Y}_0 & \mathcal{Y}_1 & \mathcal{Y}_2 & \dots \\ & \mathcal{E}'_0 \mathcal{D}_1 + \mathcal{Y}_0 & & \\ & & \mathcal{E}'_1 \mathcal{D}_2 + \mathcal{E}'_0 \mathcal{D}_1 + \mathcal{Y}_0 & \\ \vdots & \vdots & \vdots & \end{bmatrix}, \quad \mathcal{J} = \begin{bmatrix} \mathcal{Y}_0 & \mathcal{Y}_1 & \mathcal{Y}_2 & \dots \\ & \mathcal{E}'_0 \mathcal{D}_1 + \mathcal{Y}_0 & \mathcal{E}'_0 \mathcal{D}_2 & \\ & \mathcal{E}'_1 \mathcal{D}_1 & \mathcal{E}'_1 \mathcal{D}_2 + \mathcal{E}'_0 \mathcal{D}_1 + \mathcal{Y}_0 & \\ \vdots & \vdots & \vdots & \end{bmatrix}$$

- The columns of $\mathcal{J}$ (the IRFs to different current or news shocks) reflect the accumulation of terms from the fake news matrix $\mathcal{F}$.
- The algorithm requires only the computation of the primitive objects $\mathcal{Y}_t$ and $\mathcal{D}_t$ which can be obtained with one backward iteration starting from a shock at $T - 1$ and $\mathcal{E}_t$ which can be recursively obtained from $\mathbf{y}_{ss}$.
- One striking feature of the columns of the Jacobian is that they converge to a regular pattern around the main diagonal. In other words, if the shock is anticipated far enough in advance, all impulse responses are just shifted versions of each other. This reflects the fact that $\mathcal{J}_{t,s} \approx \mathcal{J}_{t-1,s-1}$ for high enough t and s (asymptotic time invariance).

Algorithm for many inputs and outputs.

$$\mathbf{Y} = h(\mathbf{X})$$

- Can be applied to each i and o , but certain objects can be reused several times.
- If $\mathcal{D}_s$ depends only on input shocks, it only needs to be computed once per input $\mathcal{D}_s^i$
- $\mathcal{E}_t$ depends only on the output of interest dY^o so we only need to compute once per output $\mathcal{E}_t^o$
- However, $\mathcal{Y}_s$ will need to be computed in for each i and o since it depends on every input dX^i and every output dY^o .

$$\mathcal{Y}_s dx \equiv dY_0^s = (d\mathbf{y}_0^s)' \mathbf{D}_{ss}$$

- $\mathcal{Y}_s^{o,i}$ are computed by doing a backward iteration in response to each input shock i , and then taking the aggregate response of each o for each s .
- Then $\mathcal{F}^{o,i}$ are computed as above but with $\mathcal{Y}^{0,i}$ in the first row and in other rows $(\mathcal{E}_t^o)' \mathcal{D}_s^i$ instead.

Notes on the backward and forward step with fake-news

For each i input in $\mathbf{X}$, shock dx for each each $s = 0, \dots, T-1$, letting $\mathbf{X}^s \equiv \mathbf{X}_{ss} + \mathbf{e}^s dx$, compute $\mathbf{y}_0^s$ and Λ_0^s as well as dY_0^s and $d\mathbf{D}_1^s$ as how these objects change from the steady state when the news hits. This is done via a single **backward iteration**. The goal is to produce all “curly” output and distributions, the first order dynamics of these objects (one is a scalar and the other a matrix)

$$\mathcal{Y}_s^i dx \equiv dY_0^s = (d\mathbf{y}_0^s)' \mathbf{D}_{ss} = (\mathbf{y}_0(\mathbf{X}^s) - \mathbf{y}_{ss})' \mathbf{D}_{ss}, \quad \mathcal{D}_s^i dx \equiv d\mathbf{D}_1^s = (d\Lambda_0^s)' \mathbf{D}_{ss} = (\Lambda_0(\mathbf{X}^s) - \Lambda_{ss})' \mathbf{D}_{ss}$$

The distribution is by far the trickiest. Recall $D_0(s, x_-) = D_{ss}(s, x_-)$

$$D_1(s', x) = \sum_{s=1} \sum_{x_-=1} \Pr(s'|s) \pi(x(s, x_-; \mathbf{X}^s)) D_0(s, x_-) = \Lambda_0(\mathbf{X}^s)' \mathbf{D}_{ss}$$

Hence, we have that only the lottery weights can change, not the distribution, which is given.

$$dD_1(s', x) = \sum_{s=1} \sum_{x_-=1} \Pr(s'|s) d\pi(x(s, x_-; \mathbf{X}^s)) D_0(s, x_-) = d\Lambda_0(\mathbf{X}^s)' \mathbf{D}_{ss}$$

These weights are coming from the optimal policies. Recall there were two weights for LHS and RHS gridpoints.

$$\begin{aligned} \pi_{ijp} D_{t-1}(s_t^i, x_{t-1}^j) &\rightarrow \tilde{D}_1(s_t^i, x_t^p) \\ \tilde{D}_1(s_t^i, x_t^{p+1}) &\leftarrow (1 - \pi_{ijp}) D_{t-1}(s_t^i, x_{t-1}^j) \end{aligned}$$

Then, since all we care about is how these weights change $d\pi_{ijp}$ to capture $d\Lambda_0(\mathbf{X}^s)$, all we need to do is set the entries to be

$$\begin{aligned} d\tilde{D}_1(s_t^i, x_t^p) &\leftarrow d\pi_{ijp} D_0(s_t^i, x_{t-1}^j) \\ d\tilde{D}_1(s_t^i, x_t^{p+1}) &\leftarrow -d\pi_{ijp} D_0(s_t^i, x_{t-1}^j) \end{aligned}$$

and apply the exogenous transition matrix, which here is invariant to shocks, to get $dD_1(s', x)$. To get the proportional change in $\frac{d\pi}{\pi_{ss}}$ we approximate it with *minus (-)* the change in the policies $-\frac{dx(s, x_{-}; \mathbf{X}^s)}{x_{ss}x(s, x_{-}; \mathbf{X})}$. This is because π is the probability of being sent down to the left gridpoint. So an increased policy function should imply a decrease in such probability.

Checking the first row and column of the Jacobian.

This will let us calculate

$$dA_1 dr^s \equiv \int a_{-} dD_1(s, a_{-}) dr^s = (\mathbf{a}_{grid})' d\mathbf{D}_1^s dr^s = \mathcal{Y}^{A,r} dr^s$$

This lets you verify that

$$\mathcal{F}_{0,:} = \begin{bmatrix} \mathcal{Y}_0 & \mathcal{Y}_1 & \mathcal{Y}_2 & \dots & \mathcal{Y}_{T-1} \end{bmatrix}$$

Fix an input in $\mathbf{X}^s$ to shock: $X^s \equiv X_{ss} + \mathbf{e}^s dx$. First, get the *contemporaneous* response $s = 0$ to unit scalar shock. Get *curlyV*, *curlyD*, *curlyY*

Fill in anticipation anticipation effects looping through $s = 1, T$.

Calculate the $(T - 1) n_g \times 1$ expectation vectors $\mathcal{E}_t \equiv (\Lambda_{ss})^t \mathbf{y}_{ss}$ from the recursion

$$\mathcal{E}_t = \Lambda_{ss} \mathcal{E}_{t-1}, \quad \mathcal{E}_0 = \mathbf{y}_{ss}$$

Sometimes this is referred to as $\mathcal{P}_u^o$ or for a given output o starting at $\mathbf{y}_{ss}^o$, the effect that a perturbation of the distribution of agents will have on aggregate output u periods ahead. Dot product of $\mathcal{D}_0^r$ and $\mathcal{P}_u^A$ lets us trace out the effect of the initial shock on aggregate assets. For example, for a change in r at 0, all we need to do is use the fact that dD_1^{r0} is $n_g \times 1$ and $\mathcal{E}_t$ is also $n_g \times 1$ to take their dot product

$$\mathcal{D}_0^r \cdot \mathcal{E}_t = dD_1^{r0} \cdot \mathcal{E}_t, \quad \forall t = 1, \dots, T$$

Doing this lets you verify that the first column of the Jacobian and the Fake News matrix are the same and can be traced out starting with the distribution change to a current $s = 0$ shock and rolled over time with

$$\mathcal{F}_{:,0} = \begin{bmatrix} \mathcal{Y}_0 \\ \mathcal{E}_0' \mathcal{D}_0 \\ \mathcal{E}_1' \mathcal{D}_0 \\ \vdots \\ \mathcal{E}_{T-2}' \mathcal{D}_0 \end{bmatrix}$$

Review of the steps to follow for fake news algorithm.

Step 0. Compute Steady State $Y(X)$ for all Y^o and X^i

Step 1. Backward and Forward Iteration to get $\mathcal{Y}_s^{o,i}$ and $\mathcal{D}_s^i$

1.1 Iterate backward *once* from a shock at $T - 1$ to get the impact on policy (Lemma 1)

→ Get $T \times n_s \times n_a$ giving us the effect on $a(s, a_-)$ for every horizon $s = 0, \dots, T - 1$

1.2 Compute *one-period-ahead* distributions from shocked asset policies (Lemma 2)

1.3 Get $\mathcal{Y}_s^{o,i}$ and $\mathcal{D}_s^i$ (for Lemma 3) calculated with “ghost runs”

$$\mathcal{Y}_s dx \equiv dY_0^s = (d\mathbf{y}_0^s)' \mathbf{D}_{ss} \implies \mathcal{D}_s dx \equiv d\mathbf{D}_1^s = (d\Lambda_0^s)' \mathbf{D}_{ss}$$

Step 2. Get $\mathcal{E}_s^o$ the expected value of each policy o at horizon s

$$\mathcal{E}_t^o = \Lambda_{ss} \mathcal{E}_{t-1}^o, \quad \mathcal{E}_0^o = \mathbf{y}_{ss}^o$$

Step 3. Form fake new matrix $\mathcal{F}$

$$\mathcal{F} = \begin{bmatrix} \mathcal{Y}_0 & \mathcal{Y}_1 & \mathcal{Y}_2 & \cdots & \mathcal{Y}_{T-1} \\ \mathcal{E}_0' \mathcal{D}_0 & \mathcal{E}_0' \mathcal{D}_1 & \mathcal{E}_0' \mathcal{D}_2 & & \mathcal{E}_0' \mathcal{D}_{T-1} \\ \mathcal{E}_1' \mathcal{D}_0 & \mathcal{E}_1' \mathcal{D}_1 & \mathcal{E}_1' \mathcal{D}_2 & & \mathcal{E}_1' \mathcal{D}_{T-1} \\ \vdots & \vdots & \vdots & & \vdots \\ \mathcal{E}_{T-2}' \mathcal{D}_0 & \mathcal{E}_{T-2}' \mathcal{D}_1 & \mathcal{E}_{T-2}' \mathcal{D}_2 & \cdots & \mathcal{E}_{T-2}' \mathcal{D}_{T-1} \end{bmatrix}$$

Step 4. Convert from fake news to Jacobians

$$\mathcal{J}_{t,s} = \begin{cases} \mathcal{J}_{t-1,s-1} + \mathcal{F}_{t,s} & t, s \geq 1 \\ \mathcal{F}_{t,s} & t = 0 \text{ or } s = 0 \end{cases}$$

IV. Examples with Jacobian implementation and dimensionality reduction techniques

We have uploaded a few jupyter notebooks. I will go over them in more detail in Friday OH.

Using 2019 SHADE/SSJ repo published version (ironed out)

1. RA RBC
2. Krusell-Smith (production economy)
3. One-Asset HANK (sticky prices)
4. Het Jacobian (under the hood of One Asset HANK) [for PSET 3]

Contain extra things: non-linear solutions, determinacy, estimation, endogenous labor supply

Using 2021 SHADE/SSJ repo beta version (not public yet, could have bugs)

1. Chapter 1 – lectures on Huggett (endowment economy)
- + methods appendix (.ipnb e.g. EGM, forward iteration)
- + standard incomplete markets model (.py)

2. Chapter 2 – lectures on canonical HANK (sticky wages)
3. Chapter 3 – lectures on fiscal policy
- + methods appendix (.ipnb on fake news algorithm)
- + fake news (.py)

V. Extensions

To be added

VI. Solving Hopenhayn Rogerson via SSJ

Recall our simplified version of Hopenhayn Rogerson with convex adjustment costs and no entry nor exit.

$$\max_n V_t(s, n_-) = \max_n \left\{ d_t + \frac{1}{1+r_t} \mathbb{E} [V_{t+1}(s', n) | s] \right\}$$

$$d_t = f(s, n_-) - w_t n_- - \Psi(n, n_-), \quad f(s, n_-) = X_t s^\alpha n_-^{1-\alpha}, \quad \Psi(n, n_-) = \frac{\psi}{2} \left(\frac{n}{n_-} - 1 \right)^2 n_-$$

We had two key distributional objects, the labor investment policies

$$n_t(s, n_-) = n(s, n_- | \{w_t, r_t\})$$

and the distribution of firms at the end of the period, evolving as

$$D_t(s', n) = \int \Pi_{s,s'} \pi(n_t^*(s, n_-) \approx n) dD_{t-1}(s, n_-)$$

Hence, since $n_t^*(\{w_t, r_t\})$ depends on the time sequence of prices, so does the distribution $D_t(\{w_t, r_t\})$.

We had two key aggregate objects, labor demand and aggregate production

$$N_t^f = \int n_- dD_t(s, n_-), \quad Y_t = \int f(s, n_-) dD_t(s, n_-)$$

Since $D_t(\{w_t, r_t\})$ depends on price sequences, so do other aggregates. Given per period investment costs, aggregate current dividends.

$$\Psi_t = \int \Psi(n^*(s, n_-), n_-) dD_t(s, n_-), \quad \Phi_t = \int d(s, n_-) dD_t(s, n_-) = Y_t - w_t N_t^f - \Psi_t$$

There were three market clearing conditions: asset, labor, and goods markets.

$$\begin{aligned} A_t &= A_t^f \\ N &= N_t^f \\ C_t &= Y_t - \Psi_t \end{aligned}$$

By Walrass law we only need two of them in every period. Given our perfect foresight dynamics, there are two no-arbitrage conditions for bonds and stocks

$$(1 + r_t) = \frac{1}{\beta} \left(\frac{C_{t+1}}{C_t} \right) = \frac{1}{\beta} \left(\frac{Y_{t+1} - \Psi_{t+1}}{Y_t - \Psi_t} \right)$$

We also have that A_t^f is defined recursively

$$A_t^f = \frac{1}{1 + r_t} \left[\Phi_{t+1} + A_{t+1}^f \right]$$

Thus, in transition, $\{w_t, r_t\}$ are pinned down to make sure that two equations hold:

$$\begin{aligned} N &= N_t^f(\{w_t, r_t; X_t\}) \\ (1 + r_t) &= \frac{1}{\beta} \left(\frac{Y_{t+1}(\{w_{t+1}, r_{t+1}; X_{t+1}\}) - \Psi_{t+1}(\{w_{t+1}, r_{t+1}; X_{t+1}\})}{Y_t(\{w_t, r_t; X_t\}) - \Psi_t(\{w_t, r_t; X_t\})} \right) \end{aligned}$$

Thus, we have that

$$\begin{aligned} H_L(\{r_t, w_t\}) &= N_t^f(\{w_t, r_t; X_t\}) - N \\ H_G(\{r_t, w_t\}) &= \frac{1}{\beta} \left(\frac{Y_{t+1}(\{w_{t+1}, r_{t+1}; X_{t+1}\}) - \Psi_{t+1}(\{w_{t+1}, r_{t+1}; X_{t+1}\})}{Y_t(\{w_t, r_t; X_t\}) - \Psi_t(\{w_t, r_t; X_t\})} \right) - (1 + r_t) \end{aligned}$$

Here we are interested, to a first order, on how N_t^f changes when any of it's inputs change at any point in time since we know that in equilibrium, it must hold that

$$\frac{\partial N_t^f}{\partial w_s} dw_s + \frac{\partial N_t^f}{\partial r_s} dr_s + \frac{\partial N_t^f}{\partial X_s} dX_s$$

$$N^f(\{w, r; X\}) - N = 0 \iff J^{N,w} dw + J^{N,r} dr + J^{N,X} dX = 0$$

Thus, clearing 2 T sets of equations with 2 T sets of unknowns (prices).

Def. Jacobians. Our goal is to calculate two Jacobians containing $\mathcal{J}^{o,i} = \left[\frac{dY_t^{(o)}}{dX_s^{(i)}} \right]_{(t,s)}$ or equivalently, that each entry $\mathcal{J}_{t,s}^{o,i} = \frac{dY_t^{s(o)}}{dx} = \frac{(Y_t^{(o)}(X_s^{(i)}) - Y_{ss}^{(o)})}{dx}$.

$$\mathcal{J}^{N^f, w} \quad \text{and} \quad \mathcal{J}^{Y, w}$$

Def. Backward iteration. Let $\mathcal{T}_w$ denote backward iteration (on the policies or equivalently on V_n or V) be denoted with assumed wage input w . That is, in the steady state *at the equilibrium* w , we have that applying one more round of backward iteration doesn't matter, since we have found a fixed point of the policy/value function.

$$\mathcal{T}_w n^*(s, n_-) \approx n^*(s, n_-)$$

Def. Forward iteration. Let $\mathcal{D}_n D$ be forward iteration that maps distribution D into next period's distribution. In the steady state, given policy n^* , we have that

$$\mathcal{D}_{n^*} D^*(s, n_-) \approx D^*(s, n_-)$$

Another way to write forward iteration is in grid representation

$$\Lambda'_{n^*} \mathbf{D}^* \approx \mathbf{D}^*$$

where $\mathbf{D}^*$ is an n_g sized vector storing the pdf of the distribution across the (s, n_-) idiosyncratic space and Λ' is the steady state transition matrix implied by policies n^* .

Let $\mathbf{n}_{\text{grid}}$ be the vector that holds the value n_- for every state (s, n_-) – i.e. the reshaped mesh.

Def. Aggregation. The aggregate labor demand N^{f*} and production at the steady state Y^* .

Def. Direct Algorithm. To compute $\mathcal{J}^{N^f, w}$ using the direct algorithm, we will iterate over T columns (horizon of the shock) and T rows (IRFs to each shock) for a given column. Recall the main steps of the direct algorithm.

- Iterate from $s = 0, \dots, T - 1$.
- For each s , take $w^s = \{w + e^s h\}$ path as given, where $h = 10^{-4}$ is small.
- Iterate *policies backwards* from $\mathbf{v}_T = \mathbf{v}_{ss}$ and compute the path of $\mathbf{v}_t^s$ for that particular shock w^s and get the associated transition matrix Λ_t^s and individual policies $\mathbf{n}_t^s$ for $t = T - 1, \dots, 0$.
- Iterate the *distribution forward* starting from $\mathbf{D}_0 = \mathbf{D}_{ss}$ applying the transition matrices Λ_t^s to solve the distributions $\mathbf{D}_t^s$ for $t = 1, \dots, T - 1$. For example,
- At each t take $N_t^s = \int n_- dD_t(s, n_-; w^s) = (\mathbf{n}_{\text{grid}})' \mathbf{D}_t^s$ using the mesh and the distribution implied by the policies, and compute numerical derivatives $\mathcal{J}_{t,s} = \frac{dY_t^s}{dx}$, **the rows of the Jacobian for each col s** , as

$$\mathcal{J}_{t,s} = \frac{dN_t^s}{dx} = \frac{(N_t(w^s) - N_{ss})}{dx}$$

- Repeat **for each of the columns** $s = 0, T - 1$, calculating the IRF for the input shock in the model.
- Do the same for $Y(w)$.

In our new notation:

- Iterate for each column $l = 0, \dots, T - 1$

1. For each l iteration, feed in a wage change at date l only. Define an input vector to be the steady state wage vector $w^s = w + h\mathbf{e}^s$, where $\mathbf{e}^s$ has only a 1 in the s 'th entry, zeros elsewhere.

2. Set the end date policy to be $n_T(s, n_-) = n^*(s, n_-)$

3. In backward iteration,

i. when $t \neq \ell$

$$n_t(s, n_-) \equiv \mathcal{T}_w n_{t+1}(s, n_-)$$

ii. when $t = l$

$$n_t(s, n_-) \equiv \mathcal{T}_{w+h} n_{t+1}(s, n_-)$$

4. Given the policies and the transition Λ_l associated with each step, roll forward the distribution $D_t, t = 0, 1, \dots, T-1$ starting from $D_0 = D^*$.

$$D_{t+1}(s, n_-) = \mathcal{D}_{n_t} D_t(s, n_-)$$

5. Compute aggregates. Note that N_t^f is a state variable in the model.

$$N_t^f \equiv \int n_- dD_t(s, n_-) = (\mathbf{n}_{grid})' \mathbf{D}_t^s$$

6. Compute the vector

$$\mathbf{v}_\ell \equiv \frac{N_t^f - N^{f*}}{h}$$

“Ghost run” Compute the steps for a single column with $h = 0$ (i.e. $w^s = w$) to only use steady state policies. Denote the resulting vector by $\mathbf{v}^0$. Set each

$$\mathbf{v}_\ell \equiv \frac{N_t^f - N^{f*}}{h} - \mathbf{v}^*$$

Def. Fake News Algorithm.

1. Set the policies at the end date to be $n_T(s, n_-) = n^*(s, n_-)$ – the steady state.

i. **Compute Expectation Vectors**

i. Use the steady state Λ transition matrix

$$\begin{aligned} \mathcal{E}_0 &= \mathbf{n}_{grid} \\ \mathcal{E}_{t+1} &\equiv \Lambda \mathcal{E}_t \end{aligned}$$

ii. $\mathcal{E}_t$ has the simple interpretation as the average level of employment a firm in a given state today expects to have t periods from now (think back to your stochastic simulation, where you applied the exogenous and endogenous components of Λ).

ii. **Perform a single Ghost Run.**

- i. Compute the T policies backward like before $n_t^*(s, n_-)$ but without feeding in a shock.
- ii. For each l denote $D_{1\ell}^* \equiv \mathcal{D}_{n_{T-1-\ell}^*} D^*$ as the distribution we arrive at after one forward iteration for each one of the n^* 's.

iii. **Perform a single Backward Iteration to get n_t^l policies and their change dn_t^l**

- i. Set $\ell = T - 1$ and compute the policies from $t = 0, T - 1$ just like we did before for a single shock happening at $\ell = T - 1$, i.e. only an MIT shock at the last date (**Lemma 1**)
- ii. This yields $n_t(s, n_-)$ for $t = 0, \dots, T - 1$, which corresponds to the policy when the wage changes in $T - 1 - t$ periods.
- iv. With the policy changes and the implied transitions, get the change in the distribution after the first row, for any $s = 0, \dots, T - 1$ compute

$$dD_s \equiv \mathcal{D}_{n_{T-1-s}} D^* - D_{1s}^*$$

where D_{1s}^* comes from the Ghost run applied to the steady state distribution with no shock, since $D_{1\ell}^* \equiv \mathcal{D}_{n_{T-1-\ell}^*} D^*$.

2. **Build the Fake News Matrix** with $\mathcal{E}_t$ and dD_s .

- i. Note that the first row of $\mathcal{J}^{N^f, w}$ and $\mathcal{F}$ is just 0's since N^f is a state variable, with $D_0 = D_{ss}$ in the steady state current employment does not change when the shock is announced.

$$\mathcal{F} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ \mathcal{E}'_0 dD_0 & \mathcal{E}'_0 dD_1 & \mathcal{E}'_0 dD_2 & & \mathcal{E}'_0 dD_{T-1} \\ \mathcal{E}'_1 dD_0 & \mathcal{E}'_1 dD_1 & \mathcal{E}'_1 dD_2 & & \mathcal{E}'_1 dD_{T-1} \\ \vdots & \vdots & \vdots & & \vdots \\ \mathcal{E}'_{T-2} dD_0 & \mathcal{E}'_{T-2} dD_1 & \mathcal{E}'_{T-2} dD_2 & \cdots & \mathcal{E}'_{T-2} dD_{T-1} \end{bmatrix}$$

3. Compute the Jacobian iteratively, as

$$\mathcal{J}_{t,s} \equiv \mathcal{J}_{t-1,s-1} + \mathcal{F}_{t,s}$$

Make sure that $\mathcal{F}$ and $\mathcal{J}$ share the same first row/col.

References

Auclert, A., Bardóczy, B., Rognlie, M. and Straub, L., 2019. Using the sequence-space Jacobian to solve and estimate heterogeneous-agent models (No. w26123). National Bureau of Economic Research. (R&R Econometrica)